TED UNIVERSITY

Syllabus for
CMPE 382 Operating Systems

Spring 2025-2026

Schedule:	
Instructors:	Sec1 & 2: Deniz Cantürk (DC), F318 Sec3 & 4 & 5: Mert Burkay Çöteli (MC)
Office Hours:	DC: Mo 10:00-11:00 MC:
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Course Code & Number	CMPE382	**Course Title**	Operating Systems
Type of Course	☑ Compulsory ☐ Elective	**Semester**	☐ Fall ☑ Spring ☐ Summer
Level of Course	BSc	**Year of Study**	Third
Course Credit Hours / ECTS	(3+0+0) 3 / 5 ECTS	**Pre-requisite / Co-requisite**	CMPE361 Computer Organization
Mode of Delivery	☑ Face-to-face ☐ Distance learning	**Language of Instruction**	☑ English ☐ Turkish
Course Coordinator	Dr. Deniz Cantürk		
Computer Usage	Linux (or Unix), C Programing language		
Textbook	Modern Operating Systems, by Andrew S. Tannenbaum, Fourth Edition, Wiley. Operating Systems Concepts by Silberschatz, Galvin and Gagne. 10th Edition, Wiley.		

Course Catalog Description	Introduction to operating systems. Historical advances. Operating system structures. The shell. System calls. Multi-processing. Multi-threading. Synchronization. Deadlocks handling. CPU scheduling. Memory managements. Storage and file systems. Input/output.
Course Objectives	The course, after introducing a historical perspective of the evolution of operating systems will cover the major components of the modern operating systems, concurrency in terms of process and thread management, CPU scheduling, synchronization, deadlock management, memory management (segmentation, paging, swapping), file systems and I/O.
Course Learning Outcomes (LO)	Upon successful completion of this course, a student will be able to 1. Recognize the organization of the computer systems. 2. Describe major components of the operating systems. 3. Implement multi-processing 4. Develop process concepts in terms of threading, scheduling, synchronization and deadlocks. 5. Manage computer memory. 6. Implement file and I/O subsystems

	TENTATIVE COURSE OUTLINE				
Week	**Dates**	**Topics**	**Learning Outcome (LO)**	**Textbook Reading**	**Assignments / Exams**
W1	9 Feb – 13 Feb	Introduction to operating systems		Ch1	
W2	16 Feb – 20 Feb	Operating system structures and hardware properties		Ch1	
W3	23 Feb – 27 Feb	Process concept		Ch2.1, 2.2	
W4	2 Mar – 6 Mar	Multi-processes		Ch2.2, 2.3	
W5	9 Mar – 13 Mar	Thread concept and multi threading		Ch2.3	HW1
W6	16 Mar – 20 Mar	Inter Process Communication- IPC		Ch2.3	
W7	23 Mar – 27 Mar	Synchronization		Ch2.3	Midterm 1
W8	30 Mar – 3 Apr	CPU Scheduling			
W9	6 Apr – 10 Apr	Deadlocks in OSs		Ch2.4	HW2
W10	13 Apr – 17 Apr	Basics of Memory management		Ch6	
W11	20 Apr – 24 Apr	Virtual memory management 1		Ch3.1, 3.2	Midterm 2
W12	27 Apr – 1 May	Virtual memory management 2		Ch3.3	
W13	4 May – 8 May	File systems		Ch 3.4, 3.7	HW3
W14	11 May – 15 May	I/O and Disk subsystem, Security		Ch 4	
W14	18 May – 22 May	Compensation for vacations during the term		Ch 5, 9	
	Note that there may be changes in the dates mentioned. Final exam date and time is to be announced later.				

Grading

A. Midterm Exam [25%]
B. Midterm Exam [25%]
C. Homework and Quizzes [15%]
D. Final [35%]

COURSE ASSESSMENTS & LEARNING OUTCOMES MATRIX	
Assessment Methods	**Course Learning Outcomes**
Assignments	LO # 1-6
Midterm Exams	LO # 1-4
Final Exams	LO # 1-6

Teaching Methods & Learning Activities	☑ Telling/Explaining ☑ Questioning ☑ Reading ☑ Demonstrating ☑ Problem Solving	☑ Collaborating ☑ Tutorial ☑ Web Searching ☐ Laboratory work
Assessment Methods (Formal & Informal)	☑ Exams ☑ Quiz	☐ Laboratory work
Student Workload (Total 150 Hrs)	☑ Readings/ Preparation ..... 63 hrs ☑ Exams / Quizzes / HWs ..... 45 hrs	☑ Lectures ..... **42** hrs ☐ Labs .....

COURSE POLICIES

I. Attendance

- Attendance is mandatory for the course.
- Students have to attend at least 18 Course Hours (17 hours of attendance will get FX grade)
- Students attended 28 Course Hours or more will be included in curving (from 27 to 18 hours of attendance will get catalogue grades)
- Check your attendance status from EduKare App, contact to your instructor immediately if there is any mistake

II. Make-up Exams

Make-ups for Midterm Exam will be available if and only if you have a legitimate reason for missing the exam (such as illness, death in family, a traffic accident, etc.). In case of an illness or emergency, you must supply a formal documentation that supports your claim.

III. Late Submission Policy

Late submissions will not be graded. There will be no make-up for homework.

IV. Cheating & Plagiarism

Collaboration is strongly encouraged; however, the work you hand in must be solely your own. Cheating and plagiarism are very serious offenses and will be penalized accordingly by the university disciplinary committee.

Cheating has a very broad description which can be summarized as "acting dishonestly". Some of the things that can be considered as cheating are the following:

- - Copying answers on exams, projects and lab works,
- - Using prohibited material on exams,
- - Lying to gain any type of advantage in class,
- - Providing false, modified or forged data in a report,
- - Plagiarizing (see below),
- - Modifying graded material to be re-graded,
- - Causing harm to colleagues by distributing false information about an exam, homework or lab.

All of the following are considered plagiarism:

- - Turning in someone else's work as your own,
- - Copying words or ideas from someone else without giving credit,
- - Failing to put a quotation in quotation marks,
- - Giving incorrect information about the source of a quotation,
- - Changing words but copying the sentence structure of a source without giving credit,
- - Copying so many words or ideas from a source that it makes up the majority of your work, whether you give credit or not.

(www.plagiarism.org)